Chenyang Zhang

+86-13122350931 | cyzhangecnu@stu.ecnu.edu.cn | IM-DM4AI

in/ chenyang-zhang-109372313 | lovelynewlife | 0009-0007-8265-8461

No. 3663 Zhongshan North Road, Shanghai - 200062, China

Brief Introduction

I am a Ph.D. candidate in Software Engineering at the School of Data Science and Engineering, East China Normal University. My research focuses on systems for AI and AI-driven system optimization, with an emphasis on execution efficiency, data movement, resource scheduling, and system-level automation. I currently study efficient AI execution in data systems, including database-native model inference, recommendation workloads, and agent-driven high-performance database optimization. Looking forward, I am interested in using AI to automate systems research workflows, including workload analysis, optimization search, implementation, and performance profiling.

Education

- **East China Normal University** — *September 2022 - Present*
  Ph.D. Candidate in Software Engineering, Data Management and Intelligence Computing (DMIC) System Group — Shanghai, China
  - GPA: 3.52/4.00, ranked in the top 5%.
  - Conducting research on systems for AI and AI-driven system optimization, advised by Prof. Chen Xu.
- **Donghua University** — *September 2018 - June 2022*
  Bachelor's Degree in Software Engineering — Shanghai, China
  - GPA: 4.13/5.00, ranked in the top 5%.
  - Outstanding Undergraduate Thesis Award.

Experience

- **Microsoft Research Asia (MSRA) []** — *May 2026 - Present*
  Research Intern — Beijing, China
  - Exploring agent-driven database engine specialization, using workload understanding and query code synthesis to generate optimized execution paths and reduce general-purpose engine overhead.
- **Transwarp []** — *October 2021 - March 2022*
  Database Development Intern — Shanghai, China
  - Participated in the development of TimeLyre, a time series database based on a fork of InfluxDB.
  - Developed a benchmark tool for time-series data ingestion, data generation, and metric measurement.
  - Analyzed user-defined functions in InfluxDB and designed an interface for UDF authorization.
  - Implemented a storage-engine optimization that improved the compression ratio for time-series data.

Selected Research Projects

- **IMBridge: [Impedance Mismatch Mitigation for Prediction Query Execution]** — *December 2023 - April 2024*
  Project Leader, in collaboration with OceanBase and Ant Group, accepted at SIGMOD/PACMMOD 2025 []
  - Identified execution-mode mismatch between database engines and AI inference frameworks.
  - Proposed an inference-aware AI function operator with runtime context reuse to avoid repeated model setup.
  - Introduced batch-aware invocation to control inference batch sizes and improve accelerator utilization.
  - Built IMBridge on OceanBase and DuckDB, achieving over 10× speedup on application workloads.
- **IMLane: [Composable Framework for Efficient AI Function Execution in Database Engine]** — *May 2024 - June 2026*
  Project Leader, in collaboration with OceanBase and Ant Group, accepted at PVLDB 2026 Industry Track []
  - Identified parallelism and scheduling bottlenecks in database-native AI function execution.
  - Designed process-level parallel execution to avoid GIL contention and enable true parallel invocation.
  - Introduced decoupled scheduling to match AI functions with heterogeneous CPU/GPU resources.
  - Built a composable C++ framework integrated with OceanBase and DuckDB, achieving 5.47× average speedup over standard OceanBase execution.

- **RECS: [Scheduling Data Processing Pipelines for Incremental Recommendation Training]** *June 2023 - February 2025*
  Core Project Member, in collaboration with Tencent Inc., accepted at SIGMOD 2025 Industrial Track
  - Identified resource idleness caused by intra-pipeline dependencies and inter-pipeline critical pipelines.
  - Proposed intra-pipeline scheduling with dynamic prefetching to overlap feature processing and embedding lookup.
  - Proposed critical-path-first inter-pipeline scheduling to reduce data processing time for incremental training.
  - Contributed to workload analysis, experiments, and paper revision; the prototype reduced training time by 20%.
- **ReTree: [Redundant Feature Test Elimination for Tree Model Inference]** *August 2025 - January 2026*
  Project Member, accepted at SIGMOD 2026 Research Track
  - Characterized redundant feature tests in decision tree and random forest inference over SQL predicates.
  - Proposed predicate-aware elimination through subtree collapse and sliding subtree reorganization.
  - Integrated the optimization into DuckDB to reduce repeated comparisons in predicate filtering.
  - Contributed to problem formulation, analysis, and revision; the prototype achieved over 2× speedup.
- **Craftsman & EncoderForge: [Efficient ML-to-SQL Translation for ML Inference Pipelines]** *July 2024 - January 2026*
  Project Member, in collaboration with PingCAP, accepted at ICDE 2025 and SIGMOD 2026
  - Investigated inefficient SQL generation for encoders in machine learning inference pipelines.
  - Designed template-based fusion rules and composite-join translation for categorical encoders.
  - Proposed cost-based selection over CASE, JOIN, and hybrid SQL translation plans.
  - Contributed to problem definition, solution design discussions, baseline experiments, and paper revision; the prototypes achieved over 2× speedup.

Ongoing Research and Development

- **AI-driven High-Performance Database Engine Optimization** *May 2026 - Present*
  Ongoing research direction
  - Exploring agent-driven database engine specialization, using workload understanding and query code synthesis to generate optimized execution paths and reduce general-purpose engine overhead.
- **SliceFlow: [Eliminating Redundant Computation for Recommendation Model Inference]** *September 2025 - Present*
  Ongoing research project
  - Studying repeated input data patterns in recommendation ranking inference, where invariant user features are recomputed across candidate items.
  - Proposing ReDR, a repeated data representation that captures redundancy levels across tensors and operators.
  - Designing operator substitution and propagation-based graph transformation rules to separate invariant and variant computation.
  - Building a Torch.fx-based prototype to capture computation graphs, apply ReDR-guided rewrites, and evaluate inference performance gains.
- **Multimodal Data Processing Engine Benchmark** *April 2025 - Present*
  Ongoing research exploration
  - Designing multimodal AI benchmark suites for Ray Data, Daft, Spark, Flink, and related data engines.
- **ZoteroOfMine: [Agent-assisted Research Workflow Toolkit]** *Janurary 2026 - Present*
  Personal research tooling project [ ]
  - Developing a Zotero-based toolkit for paper reading, local CLI utilities, clipboard protocols, and agent-assisted literature workflows.

Publications

[C.1] **Chenyang Zhang**, Junxiong Peng, Chen Xu*, Quanqing Xu, and Chuanhui Yang. (2024). IMBridge: Impedance Mismatch Mitigation between Database Engine and Prediction Query Execution. In Companion of the 2024 International Conference on Management of Data (SIGMOD).

[C.2] **Chenyang Zhang**, Junxiong Peng, Chen Xu*, Quanqing Xu, and Chuanhui Yang. (2025). Mitigating the Impedance Mismatch between Prediction Query Execution and Database Engine. In Proc. ACM Manag. Data (PACMMOD).

[C.3] Zihao Chen, **Chenyang Zhang**, Chen Xu*, Zhao Zhang, Jiaqiang Wang, Weining Qian, and Aoying Zhou. (2025). Scheduling Data Processing Pipelines for Incremental Training on MLP-based Recommendation Models. In Companion of the 2025 International Conference on Management of Data (SIGMOD).

[C.4] Qingfeng Pan, Jiahe Zhi, **Chenyang Zhang**, Chen Xu*, Zhao Zhang, Anita Shao, Guanglei Bao, Qiu Cui, Xiaowei Chen, and Aoying Zhou. (2025). Machine Learning Inference Pipeline Execution Using Pure SQL Based on Operator Fusion. In the 41st IEEE International Conference on Data Engineering (ICDE).

- **[C.5]** Mingxi Liu, Zhengyuan Ding, **Chenyang Zhang**, Qingfeng Pan, Huayou Su, Zhao Zhang, Chen Xu*, and Qingsong Ruan. (2026). Eliminating Redundant Feature Tests in Decision Tree and Random Forest Inference on SQL Predicates. In Proc. ACM Manag. Data (PACMMOD).
- **[C.6]** Qingfeng Pan, Qingyuan Jing, **Chenyang Zhang**, Jiahe Zhi, Chen Xu*, and Heng Long. (2026). EncoderForge: Generating Efficient SQL for Encoders in Machine Learning Inference Pipelines. In Proc. ACM Manag. Data (PACMMOD).
- **[C.7]** **Chenyang Zhang**, Linjun Lu, Qingfeng Pan, Chen Xu*, Xianzhong Cao, Quanqing Xu, and Chuanhui Yang. (2026). IMLane: Composable Framework for Efficient AI Function Execution in Database Engine. Accepted to PVLDB 2026 Industry Track.

Research and Technical Skills

- **Programming Languages:** Python, C++, Go, Java, Scala, JavaScript/TypeScript.
- **Systems and Frameworks:** DuckDB, OceanBase, TensorFlow, PyTorch, Torch.fx, ONNX, CUDA, Ray, vLLM, SGLang.
- **Research Skills:** Literature review, problem formulation, system prototyping, experiment design, workload analysis, performance profiling, evaluation methodology, result interpretation, and academic writing.
- **AI and Agent Tooling:** AI coding, LLM agents, tool calling, MCP, RAG, workflow orchestration, result validation.

Honors and Awards

- **ECNU Academic Innovation Promotion Program for Excellent Doctoral Students** — *July 2026*
  East China Normal University
- **SIGMOD 2025 Student Travel Grant** — *June 2025*
  SIGMOD/PODS 2025 Organizing Committee
- **SIGMOD 2024 Student Support Scholarships** — *June 2024*
  SIGMOD/PODS 2024 Organizing Committee
- **Outstanding Graduate of Shanghai** — *July 2022*
  Shanghai Municipal Commission of Education
- **Outstanding Undergraduate Thesis** — *June 2022*
  Donghua University
- **First Prize, AI Application Track** — *August 2021*
  China Undergraduate Computer Design Competition
- **Donghua University Student Scholarship** — *November 2020*
  Donghua University Education Foundation
- **Outstanding Student** — *November 2020*
  Donghua University

Volunteer Experience

- **Teaching Assistant** — *September 2024 - December 2024*
  Graduate Course: Distributed Computation Systems
  - Participated in discussions and evaluations of student projects.
  - Explored recent ideas in data processing systems through project discussions.